PHY201 Practice Exam on chapter(s): 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15

Name:____

These questions assess your understanding of the material from Chapter(s) 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15 of OpenStax College Physics. Please write your name on the sheets of paper that you use. Carefully read each question. Credit is only awarded if you answer correctly and clearly show all major steps necessary to find your answer; credit is not awarded for missing work, unclear work, or the use of memorized equations absent from the equation sheet. Half credit may be awarded for insufficient work that suggests some degree of understanding. Half credit is awarded for correct numerical answers with incorrect or missing units. Correct numerical answers do not guarantee credit.

1. Robert, a deep-sea fisherman, hooks a big fish that swims away from the boat, pulling the fishing line from the fishing reel. The whole system is initially at rest and the fishing line unwinds from the reel at a radius of 5.0 cm from the reel's axis of rotation. The reel is given an angular acceleration of 103 rad/s^2 for 2.7 s. How many meters of fishing line come off the reel in this time?

ANSWER: ________________________

2. Andrew has to push a 323-kg crate made of plasmium on a floor made of titanium over a distance of 6 m. The crate is initially at rest on the horizontal floor. What is the minimum force that Andrew must exert horizontally on the crate to start moving it? Note that for plasmium on titanium, $\mu_s = 0.53$ and $\mu_k = 0.5$.

ANSWER: ________________________

3. A kangaroo covers a distance of 89.6 m in a time of 53.2 s. What is the average speed of the kangaroo over this distance?

ANSWER: ________________________

4. The aorta pumps blood from the heart to the rest of the body. The flow rate of blood through Tyler's aorta is 11.3 L/min and Tyler's aorta has a radius of 12 mm. Recall that blood flows through smaller blood vessels known as capillaries, which have an average diameter of 8.4 μm. The speed of blood in the capillaries is about 0.28 mm/s. Calculate the number of capillaries in Tyler's blood circulatory system.

ANSWER: ________________________

5. At the end of her foot race, Samantha decelerates from a velocity of 6.6 m/s until she comes to rest, taking 1.3 s to do so. How far does Samantha travel, measuring from the end of her race until she comes to rest?

ANSWER: ________________________

6. A basketball referee tosses the ball straight up for the starting tip-off. What is the minimum velocity at which a basketball player must leave the ground in order to rise 3.19 m above the floor and make contact with the ball?

ANSWER: ________________________

7. Olivia places a 0.11-kg toy car next to a spring on a track. She moves the car backward to compress the spring 1.9 cm, then releases the car, propelling it upward along a curved track. How fast is the car going after it is released, before it begins travelling up the slope? Since Olivia greased the axles of the toy car, assume there is no friction. Olivia's previous experiments indicate that the spring has a force constant of 140 N/m.

ANSWER: ________________________

8. The magnitude of a velocity vector is 15.5 m/s and its x component's magnitude is 14.1 m/s. What is the magnitude of its y component?

ANSWER: ________________________

9. Isabella puts a coin on a scale and finds its mass is 5.77 g. When the coin and scale are completely submerged in freshwater, the scale reports the coins mass as 1.15 g. What is the density of the coin?

ANSWER: ________________________

10. Sophia inflates her bicycle tire to an absolute pressure of 0.36 MPa (a gauge pressure of about 37 psi) at a temperature of 13.2°C. What is the tire's gauge pressure after the temperature has risen to 30.7°C? Assume there are no leaks and the volume of the tire does not change.

ANSWER: ____________________________

11. Aaron normally requires 12600 kJ (about 3011 Calories) of food energy per day. If Aaron consumes 14742 kJ per day, he will steadily gain weight. How much time must Aaron spend bicycling per day, if he does not want to gain weight? Aaron's average power output while bicycling is 420 W. Report your answer in minutes.

ANSWER: ____________________________

12. Given that the planet Zephyria orbits the planet Epsilon Prime once every 141.0 days and that it is an average distance of 197,000 km from the center of Epsilon Prime, calculate the period of an artificial satellite orbiting Epsilon Prime at an average altitude of 2,370 km above Epsilon Prime's surface. Note that the radius of Epsilon Prime is 14,100 km.

ANSWER: ____________________________

13. A rocket with a mass at liftoff of 2.9 million kg burns fuel at a rate of 17500 kg/s. Its exhaust velocity was measured to be 3500 m/s. Calculate its acceleration at liftoff.

ANSWER: ____________________________

14. Mark is walking on a bridge between two mountain peaks, when curiosity suddenly overcomes him and he throws a teddy bear straight down from the bridge to the distant vale below. If the teddy bear falls 20 m in 1.3 s, then how far will it have fallen in 12.5 s?

ANSWER: ____________________________

15. Robert creates a soap bubble and measures its diameter to be 259 µm. Calculate the absolute pressure inside of the bubble, given that the surface tension of typical soapy water is 0.0370 N/m. Report your answer in mmHg, recalling that 133.32 Pa is 1 mmHg.

ANSWER: ____________________________

16. Zoe is with her daughter, Anna, at a park. She is continuously pushing on a 51-kg merry-go-round with a force of 108 N at its edge, 1.2 m away from the center. Anna watches excitedly. Calculate the angular acceleration that is produced in the merry-go-round when no one is on it. Consider the merry-go-round to be a uniform disk with negligible friction, so that its moment of inertia is $(1/2)MR^2$.

ANSWER: ____________________________

17. Dylan is designing a new 3.1-m-long, 26-kg diving board for DiveMaster, Inc. He bolted a prototype into the ground 2.5 m away from the pool-side end of the diving board with its spring being 1.9 m away from the pool-side end. Dylan notes that the spring compresses by 7 cm at equilibrium under the weight of the diving board alone. When Dylan, who has a mass of 86 kg, gets on the prototype diving board at a distance of 1.3 m from the pool-side end, by how much does the spring compress at equilibrium? Report your answer in centimeters.

ANSWER: ____________________________

18. Joshua is trying to determine the maximum mass that he can safely hold. His elbow joint is 31 mm from his bicep, 15 cm from the center of mass of his forearm, and 40 cm from the object. What is the maximum mass that Joshua can hold if, when his bicep is at its limit, his upper-arm bone exerts a force of 1000 N on his elbow joint? Note that the angle between his forearm and his upper arm is 90° and that his forearm has a mass of 2500 g. Note: the forearm is horizontal and the upper arm is vertical.

ANSWER: ____________________________

19. Elizabeth is a 60-kg athlete competing in a race. She begins from rest with an average acceleration of 1.7 m/s^2 for 2.4 s. What is her momentum at 2.4 s?

ANSWER: ____________________________

20. Nicholas wants to use his pressure cooker to cook food in liquid water at a temperature of 137°C. To determine the pressure required inside the pressure cooker for water to boil at this temperature, he can use the Clausius-Clapeyron equation for water: $P(T) = (1\,\text{kPa}) \cdot \exp\left[(4750\,\text{K})\left(\frac{1}{273.15\,\text{K}} - \frac{1}{T}\right)\right]$ In this equation, T is the temperature in Kelvin and P is the absolute pressure in the pressure cooker in kPa. Given that the lid of the pressure cooker is a disk with a diameter of 30 cm, what is the force that the lid must be able to withstand at the required pressure?

ANSWER: ____________________________

21. Nicholas flips his bicycle upside down and moves the pedals so that the rear wheel begins spinning. If the wheel starts from rest and reaches a final angular velocity of 231 rpm in 3.5 s, calculate the angular acceleration in rad/s^2.

ANSWER: ____________________________

22. Jeremy is building a seesaw for his daughter, Grace, and plans on making it 13 ft long. He plans to sit on one side, 5.9 ft away from the center of the seesaw, and his daughter will sit on the other side, 5.9 ft away from the center of the seesaw. Jeremy has weighed himself, his daughter, and the seesaw and learned that his weight is 205 lbs, his daughter's weight is 68 lbs, and the weight of the seesaw is 11.7 lbs. He wants to place the fulcrum so that the seesaw is perfectly balanced, so that Grace can cause a rotation in the seesaw just as easily as he can. Where must the fulcrum be placed?

ANSWER: ____________________________

23. Jacob is a design engineer for a lightbulb company. He manufactures a gas-filled incandescent light bulb so that the gas inside the bulb is at atmospheric pressure when the bulb has a temperature of 8°C. What is the pressure in the light bulb when it rises to 272°C? The coefficient of volume expansion for the glass light bulb is 29×10^{-6} 1/°C.

ANSWER: ____________________________

24. A typical small rescue helicopter has four blades, each of which is 4.2-m long and has a mass of 54 kg. The blades can be approximated as thin rods that rotate about one end of an axis perpendicular to their length, so that the moment of inertia of a blade is $(1/3)ML^2$. The helicopter has a total loaded mass of 1240 kg. If the helicopter's blades rotate at 332 rpm on the ground, to what height could the helicopter rise if all of its blades' rotational kinetic energy is used to lift it?

ANSWER: ____________________________

25. A typical small rescue helicopter has four blades, each of which is 3.8-m long and has a mass of 51 kg. The blades can be approximated as thin rods that rotate about one end of an axis perpendicular to their length, so that the moment of inertia of a blade is $(1/3)ML^2$. The helicopter has a total loaded mass of 970 kg. Calculate the rotational kinetic energy that the helicopter possesses when its blades rotate at 246 rpm.

ANSWER: ____________________________

26. Laika the moon dog jumps into the air to reach a height of 6.01 m. How long is Laika in the air, measured from when she jumps until she lands? Laika jumps from flat ground on the Moon and so her free-fall acceleration is 1.625 m/s^2. Note: Laika was a stray part-terrier mix from Moscow who, in 1957, became the first living creature to enter space and orbit Earth, clearing the celestial trail for human space exploration.

ANSWER: ____________________________

27. Scott, who has a mass of 103 kg, has jumped from a plane and is falling through the air at his terminal velocity, a speed of 57 m/s. How much work does the force of air resistance do against him over a distance of 9.7 m?

ANSWER: ____________________________

28. Emily drives a semi-trailer truck, which has an odometer on one hub of a trailer wheel that counts the number of wheel revolutions. The odometer calculates the distance travelled based on the rotations. If the wheel has a diameter of 1.0 m and goes through 27000 rotations, how many kilometers should the odometer read?

ANSWER: ____________________________

29. Kevin, a 71-kg ice-hockey goalie who is originally at rest, catches a 0.2-kg ice-hockey puck that is traveling at a velocity of 38.6 m/s. Calculate Kevin's speed after he catches the puck.

ANSWER: ____________________________

30. Jeremy, who has a mass of 114.1 kg, steps onto a scale that rests on the floor of an elevator. If the elevator accelerates downward at a rate of 2.5 m/s^2, what will the scale read? Explain why your answer is correct in a couple sentences.

ANSWER: ____________________________

31. Ian, a construction worker, holds up a plank of wood 3.0-m long. His left hand pushes an end of the plank down with a force F_L, while his right hand pushes the plank up with a force F_R at a distance of 0.8 m from the left hand. The plank is held in equilibrium. The plank has a mass of 18 kg and its center of gravity is located at the middle of the plank. Calculate the magnitude of force F_R.

ANSWER: ____________________________

32. Suppose that there is heat transfer of 45 J to a system, while the system does 41 J of work. Later, there is heat transfer of 50 J out of the system while 33 J of work is done on the system. What is the net change in internal energy of the system?

ANSWER: ____________________________

33. Two children, Jeremy and Scott, and a bag of marbles are balancing on a long seesaw of negligible mass. Jeremy has a mass of 21 kg and sits 4.5 m away from the fulcrum. Scott, who has a mass of 16 kg, sits on the other side of the fulcrum, opposite to Jeremy. The bag of marbles has a mass of 3.1 kg and is on Scott's side of the seesaw, a distance of 1.1 m away from the fulcrum. What is Scott's distance from the fulcrum?

ANSWER: ____________________________

34. Suppose that Jonathan floats in freshwater with 94.3% of his volume submerged when his lungs are full of air. His mass is 98 kg. What is his average density?

ANSWER: ____________________________

35. On a family trip, Charles and Leah take their son, Ian, to visit the deepest man-made hole on Earth, the Kola Superdeep Borehole. Ian shows little interest, however, and spends all his time on his cell phone. Charles, fairly horrified at his son's lack of scientific interest, grabs the phone and throws it down the hole, which is 9 inches in diameter and 12,262 meters deep. Charles manages to hear over Ian's cries that it hit the bottom in 50.9 seconds. What was the velocity at which Charles threw the cell phone?

ANSWER: ____________________________

36. Spontaneous heat transfer from hot to cold is an irreversible process. Calculate the total change in entropy if 6140 J of heat transfer occurs from a hot reservoir at 676 K to a cold reservoir at 240 K, assuming that there is no temperature change in either reservoir.

ANSWER: ____________________________

37. The aorta pumps blood from the heart to the rest of the body. Calculate the average speed of the blood in Robert's aorta if the flow rate of his blood is 7.4 L/min. Robert's aorta has a radius of 12 mm.

ANSWER: ____________________________

38. Dylan is driving in his car wondering how long would it take him, starting from 4.7 m/s and accelerating in a straight line at 4.6 m/s^2, to travel a distance of 360 m?

ANSWER: ____________________________

39. Sophia is designing a skate park and wants to build a curved ramp (quarterpipe) that leads skaters to a straight ramp. Sophia plans that skaters will drop into the quarterpipe from rest and launch off the straight ramp with a speed of 1.1 m/s. If the height of the quarterpipe is 4.8 m, what must the height of the straight ramp be? Assume that the wheel bearings are well-made and lubricated, so friction can be neglected.

ANSWER: ______________________________

40. Intravenous (IV) infusions are usually made with the help of the gravitational force. Assuming that the density of the fluid being administered is 1.00 g/mL, at what height should the IV bag be placed above the entry point so that the fluid only just enters the vein if the blood pressure in the vein is 9.2 mmHg above atmospheric pressure? Recall that 1.0 mmHg = 133.32 Pa.

ANSWER: ______________________________

41. Alexis is using a winch and rope to hold up a 6.4-kg water-filled bucket. The winch is made up of a 37-kg cylindrical log with a radius of 34 cm with a light handle extending 78 cm from the center. The rope is attached to the log, wraps around it, and is tied to the bucket. By turning the handle, Alexis would cause the log to rotate, lifting the bucket. What is the magnitude of the force that Alexis must apply to the stationary handle to keep the bucket from moving? The moment of inertia of the cylindrical log is $(1/2)MR^2$.

ANSWER: ______________________________

42. John puts a 0.58-kg aluminum pot on the stove and fills it with 0.63 liters of water. Both the pot and the water are initially at 8.4°C. How much heat is required to bring them both to 55.1°C? The specific heat value of water is 4186 J/kg°C and for aluminum it is 900 J/kg°C.

ANSWER: ______________________________

43. Two nurses, Linda and Sarah, work together to push a crash cart by pushing on the handle. They push it from rest so that it covers a distance of 10.3 m in 3.1 s. Linda exerts 140 N at 36.2° below the horizontal and Sarah exerts 140 N at 45.8° below the horizontal. The force of friction is 69 N. Determine the mass of the cart.

ANSWER: ______________________________

44. Jessica is decelerating as she snowboards at 3.4 m/s up a slope that is inclined at 3.3° above the horizontal. Calculate the magnitude of her deceleration. The total mass of Jessica and her snowboard is 88 kg. The density of the air is 1.265 kg/m^3. Note that for waxed wood on snow, $\mu_s = 0.14$ and $\mu_k = 0.1$. Ignore the effect of air resistance.

ANSWER: ______________________________

45. Daniel drives a 6000-kg truck and descends a 50-m-tall hill at a constant speed. What is the final temperature of 100 kg of brake material if its temperature is 15°C at the top of the hill? Assume the specific heat of the brake material is 810 J/kg°C. Assume the brakes do all the work to maintain the speed.

ANSWER: ______________________________

46. Stephen drives his 1740-kg car around a 430-m radius unbanked curve at 39.2 m/s (about 87 mph). Stephen's mass is 118 kg. What is the magnitude of the centripetal acceleration of Stephen's car?

ANSWER: ______________________________

47. Nicholas is spinning a large grindstone by placing his hand on its edge and exerting a force through part of a revolution. He exerts a force of 229 N through a rotation of 1.18 rad (68°). He exerts the force perpendicular to the grindstone's 58-cm radius and the effects of friction are negligible. What is the grindstone's final angular velocity if its mass is 130 kg? Consider the grindstone to be a uniform disk with negligible friction, so that its moment of inertia is $(1/2)MR^2$.

ANSWER: ______________________________

48. A 0.22-kg purple box is slid eastward on a frictionless surface into a dark room, where it strikes an initially stationary red box, which has a mass of 0.13 kg. The purple box emerges from the room at an angle of 28° north of east. The speed of the purple box is originally 7.2 m/s and is 2.9 m/s after the collision. Calculate the speed of the red box after the collision.

ANSWER: ______________________________

49. Adam places a 0.1-kg toy car next to a spring on a track. He moves the car backward to compress the spring 4.0 cm, then releases the car, propelling it upward along a curved track. How fast is the car going after it rises to a height of 0.073 cm above the starting position? Since Adam greased the axles of the toy car, assume there is no friction. Adam's previous experiments indicate that the spring has a force constant of 150 N/m.

ANSWER: ____________________________

50. Joseph drops a beanbag off the edge of a cliff and measures it striking the ground with a vertical impact velocity of 39.7 m/s. Determine the height of the cliff. Report your answer in feet, recalling that 3.28 ft = 1 m.

ANSWER: ____________________________

51. Stephen wants to place three blocks on a 6.1-kg, 19-m-long wooden board to make it balance atop a triangular wedge of wood. He places a 25-kg yellow block a distance of 1.18 m from the left end of the board, a 1.5-kg red block a distance of 14.6 m from the right end of the board, and places the board on the wedge at a distance of 3.2 m from the left end of the board. Where must he place a 1.3-kg purple block, measured from the right end of the board, in order to balance the board?

ANSWER: ____________________________

52. Patricia aims her arrow towards a target located 263.3 m away. The target's bull's eye aligns with the height from which Patricia releases the arrow. The arrow is launched at an angle with an initial speed of 95.9 m/s. Find the angle at which Patricia launched the arrow.

ANSWER: ____________________________

53. Alexis's 58-L (15.3-gal) steel gasoline tank is full of gas. Both the tank and the gasoline have an initial temperature of 11.0°C. After both the tank and the gasoline warm to 30.5°C, what is the pressure in the gasoline tank if the gasoline cannot spill out? The coefficient of volume expansion for steel is 35×10^{-6} 1/°C and for gasoline it is 950×10^{-6} 1/°C. Assume the bulk modulus B for gasoline is 1.00×10^{9} N/m^2. Show all work.

ANSWER: ____________________________

54. Calculate the work output of a Carnot engine operating between temperatures of 579 K and 403 K for 7290 J of heat transfer to the engine.

ANSWER: ____________________________

55. Kevin attaches a nozzle of diameter 6.5 mm to a garden hose of diameter 2.4 cm. If the hose fills up a 2-L bucket in 14.2 seconds, calculate the pressure in the hose assuming level, frictionless flow.

ANSWER: ____________________________

56. Robert is trying to keep a 30.4-kg wagon at rest on a frictionless inclined plane, but despite his best efforts, he can only apply a force of 40 N to try to stop the motion. If the plane makes an angle of 16.4° with the horizontal, what is the acceleration of the wagon down the plane?

ANSWER: ____________________________

57. Ashley is designing a skate park and wants to build a curved ramp (quarterpipe) that leads skaters a distance of 5.4 m to the top of a straight ramp. Ashley plans that skaters will drop into the quarterpipe from rest and launch off the straight ramp with a speed of 0.5 m/s. If the height of the quarterpipe is 2.6 m, what must the height of the straight ramp be? Experimental studies in skate science show that the average friction force per mass on skaters as they skate across this particular surface is 0.46 N/kg.

ANSWER: ____________________________

58. A particular wildebeest accelerates from rest to a speed of 7.1 m/s in 4.9 s. Calculate its acceleration.

ANSWER: ____________________________

59. Olivia is with her daughter, Sarah, at a park. Sarah has climbed onto a merry-go-round. Olivia is continuously pushing on the 34-kg merry-go-round with a force of 180 N at its edge, 1.7 m away from the center. Calculate the angular acceleration that is produced in the merry-go-round while 25-kg Sarah is 1.19 m away from its center. Consider the merry-go-round to be a uniform disk with negligible friction, so that its moment of inertia is $(1/2)MR^2$.

ANSWER: ______________________________

60. Victoria places an outdoor sandwich-board advertising sign on the ground. The sign used to have a chain 0.43 m vertically below its hinge, but the chain is broken. The sign's mass is 4 kg. The sign has a vertical height of 1.65 m. The hinge is opened so that the feet of the sign measure 2.7 m apart. What is the minimum coefficient of static friction between the feet of the sign and the sidewalk? To draw the sign, you may draw a large A. The chain, now broken, would have resembled the horizontal line in the A and the hinge is at the top of the A.

ANSWER: ______________________________

61. Rachel goes outside on a summer morning with her tea and sees a perfectly vertical strand of spider web. She watches for a while and eventually a spider climbs up the strand. At this point, invested in her experiment, Rachel takes the spider and puts it on a scale, measuring its mass to be 382 mg, then returns it to its vertical strand of web. What is the tension in the vertical strand of web when the spider hangs from it?

ANSWER: ______________________________

62. Patricia is watching her dog, Zeus, play around in the backyard. Zeus is standing still until he sees a squirrel. Over 4.7 s, Zeus runs a distance of 12.7 m in a straight line. Patricia is curious: how fast was Zeus moving at 4.7 s, assuming that his acceleration was constant?

ANSWER: ______________________________

63. Ian is a contestant in a winter sporting event. He is tasked with pulling a rope connected to a 310-kg block of ice on a frozen lake. Ian pulls diagonally upward on the rope at an angle of 26° from the vertical. What is the minimum force with which he must pull in order to get the block of ice moving? Note that for ice on ice, $\mu_s = 0.1$ and $\mu_k = 0.03$.

ANSWER: ______________________________

64. Alexis throws a brick horizontally from her dorm window 18.2 m above the ground. Stephen, another student, stands 11.0 m away from the dorm building and catches the brick at a height of 1.71 m above the ground. Stephen claims that Alexis threw the brick too slowly, so Alexis wants to calculate its initial speed to prove him wrong. Ignoring friction, what initial speed does she calculate?

ANSWER: ______________________________

65. A man walks 158.6 m east and then 139.5 m south. What is the difference between his travelled distance and the absolute value of his displacement?

ANSWER: ______________________________

66. Tyler shoots his arrow at a target 204.4 m away. The target's bull's eye is level with the arrow's release point. Launched at an angle and with an initial speed of 70.5 m/s, the arrow passes by a tree branch, which is 104.2 m from Tyler and 13.0 meters above the launch point, and buries itself into the bull's eye. As the arrow passes by the branch, what is its vertical displacement relative to the branch? In other words, how far above or below does it pass the branch?

ANSWER: ______________________________

67. James's doctor performs a special test and tells him that a 2.8-N force can rupture his eardrum. If his eardrum has an area of 1.04 cm^2, at what depth in freshwater would his eardrum rupture, assuming the gauge pressure in the middle ear is zero and he does nothing to alleviate the pressure?

ANSWER: ______________________________

68. A particular hydraulic system contains two fluid-filled cylinders which are connected by a tube. Cylinder 1 has a diameter of 1.2 cm and is capped with piston P_1. Cylinder 2 has a diameter of 5.0 cm and is capped with piston P_2. A force F_1, equal to 80 N, is exerted on P_1. What is the magnitude of the resulting force exerted by P_2?

ANSWER: ____________________________

69. Christopher, a 77.3-kg sprinter, starts a 800-m race from rest and reaches the 3-m mark in 1.6 s. What is his average net force that he exerts on the ground, propelling himself forward?

ANSWER: ____________________________

70. Grace is riding a motorcycle that can produce an acceleration of 5.0 m/s^2 while travelling at 90 km/h. The magnitude of the two forces resisting motion, friction and air resistance, total 370 N. The mass of Grace and the motorcycle is 330 kg. What is the magnitude of the force that the motorcycle's rear tire exerts on the ground?

ANSWER: ____________________________

71. Two billiard balls, specifically the one-ball and the thirteen-ball, move toward each other until they elastically collide. The thirteen-ball begins at rest until a force of 32 N is applied to it for a brief 11 ms, at which point it is moving toward the one-ball. Calculate the velocity of the one-ball after it elastically collides with the thirteen-ball, given that each ball has the same mass of 157 g and the velocity of the one-ball is -3.9 m/s, moving toward the thirteen-ball.

ANSWER: ____________________________

72. Calculate the depth below the surface of freshwater at which the pressure due to the weight of the water equals 1.4 atm.

ANSWER: ____________________________

73. Calculate the maximum buoyant force that saltwater could exert on a ship with a mass of 2.1 $\times$ 10^8 kg if the ship displaces 9.5 $\times$ 10^6 m^3 of water. Saltwater has a density of 1025 kg/m^3.

ANSWER: ____________________________

74. Joseph is spinning a large grindstone by placing his hand on its edge and exerting a force through part of a revolution. How much work does he do if he exerts a force of 256 N through a rotation of 0.95 rad (54°)? He exerts the force perpendicular to the grindstone's 51-cm radius and the effects of friction are negligible. Consider the grindstone to be a uniform disk with negligible friction, so that its moment of inertia is $(1/2)MR^2$.

ANSWER: ____________________________

75. When rebuilding his car's engine, Kevin finds that he must exert 260 N of force to insert a dry steel piston into a steel cylinder. What is the magnitude of the normal force between the piston and cylinder as the piston is inserted into the cylinder at a constant speed? Note that for dry steel on dry steel, $\mu_s = 0.6$ and $\mu_k = 0.3$.

ANSWER: ____________________________

76. A powerful motorcycle can accelerate from rest to 35.8 m/s in 4.2 s. What is the angular acceleration of its 0.34-m-radius wheels?

ANSWER: ____________________________

77. A freshwater reservoir has a surface area of 66 km^2 and an average depth of 11 m. What mass of water is held behind the dam?

ANSWER: ____________________________

78. Kevin is hanging from a pull-up bar inside an elevator that is accelerating downward at a rate of 6.2 m/s^2. Kevin's mass is 89.8 kg and the distance between the elevator floor and the bottom of his feet is 0.79 m. Kevin lets go of the bar just as the elevator changes its acceleration to 5.1 m/s^2 downward. At what speed will he hit the elevator floor?

ANSWER: ____________________________

79. What is the average kinetic energy of a gas molecule at 60°C?

ANSWER: ____________________________

80. Elizabeth is skiing at a speed of 12.2 m/s just as she crests a hill. On the downslope, she accelerates at 3.5 m/s^2 for 7.5 s. Calculate her speed at the end of the 7.5 s.

ANSWER: ____________________________

81. Victoria is producing heat at a rate of about 239 W while doing light exercise. At what rate must water evaporate from her body to get rid of the produced energy? The latent heat of vaporization of water is 2430 kJ/kg at body temperature, 37°C.

ANSWER: ____________________________

82. Andrew is playing tennis when he throws a 48-g tennis ball up in the air, hitting it when it's at its peak to give it a speed of 49.9 m/s, 2.0° above the horizontal. What is the magnitude of the average force that Andrew's raquet exerts on the tennis ball, given that the ball remained in contact with the racquet for 4.6 ms?

ANSWER: ____________________________

83. William drives his 1320-kg car around a 360-m radius unbanked curve at 34.5 m/s (about 77 mph). William's mass is 83 kg. Find the minimum static coefficient of friction between the tires and the road that prevents the car from slipping.

ANSWER: ____________________________

84. A particular dam is 210-m wide and contains freshwater that is 12-m deep. What is the average pressure on the dam due to the water?

ANSWER: ____________________________

85. Scott is at the range firing his M16A2 rifle. When firing a standard 5.56x45mm NATO cartridge, the velocity at which the bullet leaves the muzzle is 3100 ft/s. If Scott were to fire 23.8° above the horizontal, what maximum height would the bullet reach? Report your answer in feet. Ignore air resistance and friction in the barrel. Free-fall acceleration in English units is 32.2 ft/s^2.

ANSWER: ____________________________

86. A cricket just barely jumps over a 1.62-m-tall bush. How long is the cricket in the air, measured from when it jumps to when it lands?

ANSWER: ____________________________

87. To test the idea of surviving a free-falling elevator, Jessica conducts an experiment using a string-attached plank and a programmable toy flea. Her makeshift elevator falls 18.4 m and the flea jumps 0.0001 s before impact. Previous tests show that these toys can endure impacts below 6.74 m/s. What is the minimum initial speed needed for the flea's vertical jump to make it impact the ground at a safe speed? (Note: The plank's mass is significantly greater than the toy's mass.)

ANSWER: ____________________________

88. Victoria is an astronaut walking on the surface of the planet Xanadu. She experiences a small acceleration due to Earth's gravitational field, even at the distance between the center of Earth and Victoria of 0.28 Gm. What is her acceleration due to the force exerted by Earth's gravitational field alone?

ANSWER: ____________________________

89. Dylan is sitting in his folding chair on the beach, watching his friend Ryan as he rides in a kayak moving at a velocity of 5.23 m/s at an angle of 50.8° east of north. Ryan then passes his sister, Sarah, who is riding in a submarine, which, from Ryan's perspective, is travelling at a velocity of 1.03 m/s at an angle of 33.7° south of east. From Dylan's perspective, in what direction is Sarah's submarine travelling?

ANSWER: ____________________________

90. John's 62-L (16.4-gal) steel gasoline tank is full of gas. Both the tank and the gasoline have an initial temperature of 14.7°C. How much gasoline has leaked out due to thermal expansion by the time both the tank and the gasoline have warmed to 30.2°C? The coefficient of volume expansion for steel is 35×10^{-6} 1/°C and for gasoline it is 950×10^{-6} 1/°C. Show all work.

ANSWER: ______________________________

91. Sarah and Brandon are on a slowly sinking ship, but they have hope: Brandon managed to find an oaken door that is 219-cm tall, 110-cm wide, and 5-cm deep. Perhaps they could both fit on the door if he throws it into the sea. What is the maximum mass the door will support? Sarah recently took a physics class and remembers the density of oak is about $790\ \mathrm{kg/m^3}$ and the density of seawater is $1025\ \mathrm{kg/m^3}$.

ANSWER: ______________________________

92. Joshua, a fireman by profession but physicist at heart, jumps horizontally from a burning building with an initial speed of 7.8 m/s. Later, Joshua is interested in determining the time at which the angle between his velocity and acceleration vectors is 36°. At what time does that happen?

ANSWER: ______________________________

93. Sarah attaches a nozzle of diameter 4.5 mm to a garden hose of diameter 1.9 cm. If the hose fills up a 3-L bucket in 13.1 seconds, what is the speed of the water that leaves the nozzle?

ANSWER: ______________________________

94. While Paul is fishing, he begins to reel in a reel big fish. His "FishFinder 3000" tells him that this particular fish has a volume of $9500\ \mathrm{cm^3}$. As he's reeling the fish in, he notes that his reel ($I = MR^2/2$) begins at rest and revolves 8 times in 11 seconds. The 500-g reel has a radius of 4 cm and a friction torque of 35 N cm. How much force did Paul apply to the reel's crank handle to reel the fish in if the handle is a distance of 6.4 cm from the center of the reel and the fish was pulled vertically upwards through saltwater, which has a density of $1025\ \mathrm{kg/m^3}$? Later measurements of the fish reveals it had a mass of 10.95 kg. You may ignore the effects of the drag force.

ANSWER: ______________________________

95. In a particular year, the main span of San Francisco's Golden Gate Bridge was 1273-m long on the coldest day. The bridge was exposed to temperatures ranging from -9°C to 35°C. What is the change in its length between these temperatures? Assume that the bridge is made entirely of steel so that its coefficient of linear expansion is 12×10^{-6} 1/°C.

ANSWER: ______________________________

96. Aaron is playing baseball and is sliding into second base. He has a mass of 75.5 kg and begins sliding at a speed of 5.5 m/s. The average friction force acting against him during his motion is 730 N. What is the distance that he slides? Full credit will not be awarded if you fail to consider the signs (+/-) of quantities correctly.

ANSWER: ______________________________

97. A team of five huskies pulls Jason and his sled across snow at 3.7 m/s. Jason and his loaded sled have a combined mass of 310 kg and the average mass of each dog is 25.2 kg. If each dog can pull with a force of 140 N, what is the magnitude of the acceleration of the sled? Note that for waxed wood on snow, $\mu_s = 0.14$ and $\mu_k = 0.1$.

ANSWER: ______________________________

98. Benjamin throws a 3.2-g coin straight downward from a 30-story window with a vertical speed of 8.3 m/s (approximately 18.6 mph). Ignoring the effect of the drag force, at what speed does the coin impact a patch of grass on the ground? Express the speed in meters per second. Take a story as roughly 10.3 feet. There are 3.28 feet in one meter.

ANSWER: ______________________________

99. If a marathon runner averages 94 furlongs per deciday, then how long would it take her to run a 6.4-league marathon? Report your answer in minutes. A furlong is 10 chains, a mile is 1760 yards, a chain is 22 yards, a day is 10 decidays, and a league is 3 miles.

ANSWER: ______________________________

100. Out of curiosity, you want to conduct some geographic measurements of your house and your friend Victoria's. To get to Victoria's, you have to walk 3.69 km west and then 2.71 km north. What is the angle of a line connecting your house to hers?

ANSWER: ______________________________

101. Thomas observes a 404-g squirrel with a cross-sectional area of 326 cm^2 falling from a large tree to the ground. The temperature of the air is 19.4°C. Recall that the density of air ρ (in kg/m^3) depends on temperature T (in °C) according to $\rho = \frac{353}{T+273}$. Assume the squirrel's drag coefficient of 1.0. Assuming the squirrel reaches terminal velocity, at what speed will it hit the ground?

ANSWER: ______________________________

102. A 59.5-kg block is resting on a so-called "heavy-duty" shelf at a height of 5.6 m above the ground. The shelf suddenly collapses and the block falls to the ground. What is the block's total mechanical energy at a height of 3.9 m, assuming negligible air resistance?

ANSWER: ______________________________

103. Ethan is not only a part-time guide at the Battleship USS Iowa Museum, but also a physics student. He wants to find the net external force that would be exerted on a 1120-kg artillery shell fired from one of the USS Iowa's turrets. Ethan measures the barrel of a turret to be 19.9-m long and reads that a shell, fired from rest, would leave the barrel at 660 m/s. What is the net external force exerted on the shell?

ANSWER: ______________________________

104. William kicks a soccer ball, initially at rest on the ground, with an initial velocity of 34.0 m/s at a launch angle of 50.2° above the horizontal. The ball flies through the air and lands back on the ground. Calculate its total flight time, assuming that air resistance is negligible.

ANSWER: ______________________________

105. Adam places a 11.4-kg red wooden block on a wooden table. He then attaches a pulley to the table, threading a rope through the pulley to a 7.5-kg blue block. This blue block is hanging in the air by the rope and does not make contact with anything. What is the acceleration of the blue hanging block? Note that for wood on wood, $\mu_s = 0.5$ and $\mu_k = 0.3$.

ANSWER: ______________________________

106. Find the increase in entropy as 3.2 kg of ice, originally at 0.5°C, melts to form water at 0.5°C at a particular pressure. The latent heat of fusion for water is 334 kJ/kg.

ANSWER: ______________________________

107. Grace is an astronaut working outside the International Space Station (ISS) where the atmospheric pressure is essentially zero. The pressure gauge on the astronaut's air tank reads 7.2×10^6 Pa. What force does the air inside the tank exert on the flat end of the cylindrical tank, a disk that is 0.24 m in diameter?

ANSWER: ______________________________

108. A great blue heron is flying horizontally at a speed of 5.0 m/s when the herring in her talons wiggles loose and falls into the lake 16.1 m below. Calculate the speed at which the fish hits the water.

ANSWER: ______________________________

109. A BASE jumper runs off a cliff. When he loses contact with the cliff, he is traveling purely in the x-direction at a speed of 2.2 m/s. What is his overall speed after 1.25 seconds?

ANSWER: ______________________________

110. A 2.49-mg flea jumps into the air by exerting a force straight down on the ground. A breeze blows on the flying flea and exerts a horizontal force of 7 μN on the flea. Find the magnitude of the acceleration of the flea while it is airborne.

ANSWER: ______________________________